INVARIANTS OF A FINITE GROUP ACTION MODULO INVARIANT IDEAL GIVE AN INTEGRAL UNIVERSALLY BIJECTIVE EXTENSION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Polynomial lifting: every polynomial with coefficients in (R/I)^G lifts to R^G. invoked at proof lines 32563-32565. Verifies the hypothesis under which the universally bijective lemma applies to the map R^G/I^G into (R/I)^G.
- A ring map satisfying the universally bijective hypothesis is integral, induces a homeomorphism on spectra and purely inseparable residue field extensions. invoked at proof lines 32563-32566. Provides the conclusion of the lemma.

Source reference. Original lemma 0BRF: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32540–32540. Cited at proof lines 32563-32565 to verify the hypothesis of the universally bijective lemma; load-bearing inside the counted span.

Source reference. Original lemma 09EF: sources/stacks-chow/tags.txt, source lines 11898–11898. Cited at proof lines 32563-32566 to supply the conclusion once its hypothesis is verified; the conclusion is the statement being proved.

Lemma 0.2. *Let R be a ring. Let G be a finite group acting on R. Let $I \subset R$ be an ideal such that $\sigma(I) \subset I$ for all $\sigma \in G$. Then $R^G/I^G \subset (R/I)^G$ is an integral extension of rings which induces a homeomorphism on spectra and purely inseparable extensions of residue fields.*

Proof. Since $I^G = R^G \cap I$ it is clear that the map is injective. Lemma 0BRF shows that Algebra, Lemma 09EF applies. □

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